SAT Math Desk Packet

Student: Karina
Goal: Digital SAT Math about 680 → about 780
What this is: Math notes and exam-style questions only. No lesson clock. Use it at the desk: point at a skill, do the items, check the back.

Items are **instructor-created** in Digital SAT style (short stem, 4-choice MC or SPR). They are not copied from Bluebook. Official practice stays in Bluebook / the Student Question Bank.

PSDA = Problem-Solving and Data Analysis (ratios, percents, data, probability, statistical claims). About 15% of Math.

Answer keys are the last section. Do not flip there during a timed set.

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1. How Digital SAT Math is built

Fact	Detail
Time	2 modules × 35 minutes, 22 questions each (44 questions, 70 minutes)
Calculator	Allowed on **every** Math question (Desmos is built into Bluebook)
Formats	About 75% multiple choice (A–D); about 25% student-produced response (SPR)
Adaptive	Module 1 routes Module 2. Easy Module 1 misses can cap a 780
Mix	Every module includes all four domains below

Domain	Share	Typical count	What SAT actually asks
A Algebra	~35%	13–15	Linear equations, linear functions, systems, inequalities
B Advanced Math	~35%	13–15	Equivalent expressions; quadratic / exponential / abs / rational / radical; nonlinear systems
C PSDA	~15%	5–7	Ratios, rates, units, percents, tables, scatterplots, probability, margin of error, study claims
D Geometry & Trig	~15%	5–7	Area and volume, similar triangles, Pythagorean, special rights, $\sin \cos \tan$, circles

SPR: type a number (integer, decimal, or fraction). Blank SPR is a free miss.

2. Math notes (what to have in hand)

Algebra

- Isolate: expand → combine like terms → divide. If you see $5 - 2(3x + 1)$, the minus hits **both** terms inside.
- Fractions: multiply **every** term by the LCD.
- Slope: $m = \frac{y_2 - y_1}{x_2 - x_1}$. Parallel = same m. Perpendicular = negative reciprocal.
- $y = mx + b$: m is the rate (units of y per 1 unit of x); b is the value when $x = 0$.
- System: 1 solution (intersect), 0 (parallel distinct), infinitely many (same line).
- Inequality: flip **only** when multiplying or dividing by a negative.

Advanced Math

- "Equivalent" = same value for all allowed x. Factor a common binomial, or expand, or graph both.
- Quadratic $ax^2 + bx + c$: zeros from factors; vertex $x = -\frac{b}{2a}$; vertex form $a(x - h)^2 + k$ has vertex (h, k).
- If $a > 0$, minimum is k. If $a < 0$, maximum is k.
- Exponential: $y = A \cdot r^x$. $A = y(0)$. Growth $r > 1$; decay $0 < r < 1$. "+20%" means $\times 1.20$, not $+20$.
- $|A| = k$: if $k < 0$, no solution; if $k = 0$, one; if $k > 0$, two cases.
- Rational: multiply by LCD; throw away any x that makes a denominator 0.
- Radical: isolate, square, **check** in the original (extraneous roots are a real SAT skill).

PSDA

- Ratio $a : b$ with first part A: $\frac{a}{b} = \frac{A}{B}$.
- Combined rate: add the rates, then time $= \frac{\text{work}}{\text{combined rate}}$.
- Percent of: part $= p\times$ whole. Percent change $= \frac{\text{new} - \text{old}}{\text{old}}$.
- Successive percents multiply: up 10% then up 10% is $\times 1.1 \times 1.1 = 1.21$, not $+20\%$.
- "30% of the 40% that are red" is $0.3 \times 0.4 = 0.12$, not $40 - 30$.

- Mean follows outliers; median often does not. Mean of n values $\Rightarrow$ sum $= n \cdot \text{mean}$.
- Two-way table: $P(A)$ uses the grand total; $P(A \mid B)$ first restrict to the B row or column.
- Margin of error: 52% $\pm$ 4% means the true value is **likely between 48% and 56%**, not "exactly 52%."

Geometry and trigonometry

- Linear scale $k \Rightarrow$ area $\times k^2$, volume $\times k^3$.
- Similar figures: sides and perimeter $\times k$; **angles stay the same**.
- Triangle angles sum to 180°.
- Pythagorean: $a^2 + b^2 = c^2$. Families: 3-4-5, 5-12-13, 6-8-10, 9-12-15.
- 45-45-90: legs s, hypotenuse $s\sqrt{2}$.
- 30-60-90: short s (opp. 30°), long $s\sqrt{3}$, hypotenuse $2s$.
- $\sin\theta = \dfrac{\text{opp}}{\text{hyp}}$, $\cos\theta = \dfrac{\text{adj}}{\text{hyp}}$, $\tan\theta = \dfrac{\text{opp}}{\text{adj}}$. Also $\cos\theta = \sin(90^\circ - \theta)$.
- Circle: $C = 2\pi r = \pi d$, $A = \pi r^2$, arc $= \dfrac{\theta}{360} \cdot 2\pi r$, sector $= \dfrac{\theta}{360} \cdot \pi r^2$.
- $(x-h)^2 + (y-k)^2 = r^2$: center (h, k), radius r.

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3. Algebra questions

Short stems, SAT phrasing. MC unless marked SPR. Desmos allowed.

A1. The functions f and g are defined by $f(x) = x + 7$ and $g(x) = 7x$. What is the value of $4f(2) - g(2)$?
A) -5 B) 9 C) 22 D) 36

A2. The graph of $5x + 2y = 10$ in the xy-plane is a line. What is the slope of the line?
A) $-\frac{5}{2}$ B) $-\frac{2}{5}$ C) $\frac{2}{5}$ D) $\frac{5}{2}$

A3. The cost C, in dollars, of an order of n shirts is $C = 12n + 45$. Which statement is the best interpretation of the number 12 in this equation?
A) The cost of 12 shirts
B) The cost, in dollars, of each additional shirt
C) The fixed fee, in dollars
D) The number of shirts that cost \$45

A4. A gym charges a \$40 signup fee plus \$25 each month. After m months the total paid is \$165. Which equation represents this relationship?
A) $40m + 25 = 165$ B) $40 + 25m = 165$ C) $40 + 165m = 25$ D) $25 + 165m = 40$

A5. What is the solution to the system $x + y = 10$ and $2x - y = 5$?
A) $(5, 5)$ B) $(4, 6)$ C) $(6, 4)$ D) $(3, 7)$

A6. (SPR) Adult tickets cost \$12 and child tickets cost \$7. A group buys 15 tickets for \$145. How many adult tickets did they buy?

A7. How many solutions does the system $2x - 4y = 6$ and $x - 2y = 5$ have?
A) Zero B) Exactly one C) Exactly two D) Infinitely many

A8. Which values of x satisfy $-2x + 5 > 13$?
A) $x > -4$ B) $x < -4$ C) $x > 4$ D) $x < 4$

A9. Which point (x, y) is a solution to both $y \le -x + 2$ and $y \ge 0$?
A) $(3, 0)$ B) $(0, 3)$ C) $(1, 0.5)$ D) $(-1, 4)$

A10. A line in the xy-plane passes through $(2, 5)$ and $(6, 13)$. Which equation represents the line?
A) $y = 2x + 1$ B) $y = 2x - 1$ C) $y = \frac{1}{2}x + 4$ D) $y = 8x - 11$

A11. Which equation represents the line that is perpendicular to $y = 3x - 1$ and passes through $(0, 4)$?

A) $y = 3x + 4$ B) $y = -\frac{1}{3}x + 4$ C) $y = \frac{1}{3}x + 4$ D) $y = -3x + 4$

A12. (SPR) $\frac{5x-2}{4} = \frac{x+6}{2}$. What is the value of x?

A13. (SPR) A tank contains $W = 120 - 4t$ liters of water after t minutes. After how many minutes is the tank empty?

A14. For which values of k is $x = 2$ a solution of $3x + k > 10$?

A) $k > 4$ B) $k < 4$ C) $k > 16$ D) $k = 4$

A15. What is the solution of $7 - 3(2x - 1) = 2x + 8$?

A) $x = \frac{1}{4}$ B) $x = 0$ C) $x = 1$ D) $x = 2$

A16. (SPR) A phone plan costs \$35 per month plus \$0.10 per text. The total bill is \$47. How many texts were sent?

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4. Advanced Math questions

B1. Which expression is equivalent to $2x(x+4)-(x+4)$?
A) $(x+4)(2x-1)$ B) $(x+4)(2x+1)$ C) $2x^2+7x-4$ D) $2x^2+8x-4$

B2. Which expression is equivalent to $\dfrac{x^2-9}{x-3}$ for $x \neq 3$?
A) $x-3$ B) $x+3$ C) x^2-3 D) $9-x$

B3. (SPR) $(3x^2y)^2 = ax^by^c$, where a, b, and c are positive. What is the value of $a+b+c$?

B4. Which expression is equivalent to $8x^3-27$?
A) $(2x-3)^3$ B) $(2x-3)(4x^2+6x+9)$ C) $(2x-3)(4x^2-6x+9)$ D) $(8x-27)(x^2+1)$

B5. The function f is defined by $f(x)=2(x-3)(x+1)$. What are the zeros of f?
A) 3 and -1 B) -3 and 1 C) 2 and 0 D) 6 and -2

B6. The function g is defined by $g(x)=x^2+55$. What is the minimum value of $g(x)$?
A) 3025 B) 110 C) 55 D) 0

B7. (SPR) For $y=x^2-8x+10$, what is the x-coordinate of the vertex of the graph in the xy-plane?

B8. (SPR) What is the maximum value of $y=-(x+2)^2+9$?

B9. What are the solutions of $|3x+1|=10$?
A) 3 and -3 B) 3 and $-\dfrac{11}{3}$ C) $\dfrac{11}{3}$ and -3 D) 3 only

B10. How many distinct real solutions does $|x-2|=-5$ have?
A) Zero B) One C) Two D) Infinitely many

B11. (SPR) What is the solution of $\dfrac{x+3}{x-1}=2$?

B12. Which is a valid solution of $\sqrt{2x+3}=x$?
A) -1 only B) 3 only C) -1 and 3 D) There are no valid solutions

B13. What is the solution of $2^{x+1}=16$?
A) $x=2$ B) $x=3$ C) $x=4$ D) $x=8$

B14. (SPR) A population starts at 200 and grows by 50% each decade. What is the population after 2 decades?

B15. In the xy-plane, how many intersection points do $y=x^2$ and $y=2x+3$ have?
A) 0 B) 1 C) 2 D) 3

B16. (SPR) The function f is defined by $f(x) = 3 \cdot 2^x$. What is $f(4)$?

5. PSDA questions

C1. (SPR) A mix uses flour to sugar in the ratio $2 : 5$. If 12 cups of flour are used, how many cups of sugar are needed?

C2. (SPR) A car travels 180 miles in 3 hours. At that constant rate, how many miles does it travel in 2.5 hours?

C3. In a group of items, 40% are red. Of the red items, 30% also have stripes. What percent of all the items are red and have stripes?
A) 10% B) 12% C) 70% D) 75%

C4. A price of \$80 increases by 25% and then the new price decreases by 25%. What is the final price?
A) \$80 B) \$75 C) \$85 D) \$70

C5. (SPR) 18 is 45% of what number?

C6. A population of 800 grows by 10%, then grows by 10% again. What is the new population?
A) 960 B) 968 C) 880 D) 1000

C7. (SPR) What number, increased by 20%, equals 72?

C8. For the list $2, 2, 3, 7, 16$, what is the median?
A) 2 B) 3 C) 6 D) 7

C9. (SPR) Karina’s mean on five tests is 80. Four of the scores are 72, 88, 90, and 70. What is the fifth score?

C10. Club members:

	Yes	**No**	**Total**
Junior	12	8	20
Senior	18	12	30
Total	30	20	50

If a member is selected at random from the **seniors**, what is the probability the member said Yes?
A) $\frac{18}{50}$ B) $\frac{18}{30}$ C) $\frac{30}{50}$ D) $\frac{18}{20}$

C11. A random sample of voters shows 52% support, with a margin of error of 4%. Which conclusion is most appropriate?

A) Exactly 52% of all voters support the candidate.

B) The true support is likely between 48% and 56%.

C) The candidate will win.

D) The sample is biased because the result is not 50–50.

C12. A researcher asks only students in the honors hallway whether they like a new schedule, then claims the **whole school** supports it. The main problem with this claim is:

A) The mean is undefined.

B) The sample is not representative of the school.

C) The margin of error must be 0.

D) This is a well-designed experiment.

C13. The density of a wood is 8 grams per cubic centimeter. A sample has mass 40 grams. What is the volume of the sample, in cubic centimeters?

A) 5 B) 32 C) 48 D) 320

C14. Using the club table in C10, what is $P(\text{Senior and Yes})$?

A) $\frac{18}{30}$ B) $\frac{18}{50}$ C) $\frac{30}{50}$ D) $\frac{18}{20}$

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6. Geometry and trigonometry questions

D1. (SPR) What is the volume of a cube with edge length 3?

D2. Every edge of a cube is multiplied by $\frac{1}{2}$. The new volume is what fraction of the original volume?
A) $\frac{1}{2}$ B) $\frac{1}{4}$ C) $\frac{1}{8}$ D) $\frac{1}{16}$

D3. Two similar paintings have corresponding side lengths in the ratio $2:3$. The smaller painting has area 20. What is the area of the larger painting?
A) 30 B) 40 C) 45 D) 60

D4. A triangle has angles measuring 40° and 65°. What is the measure of the third angle?
A) 65° B) 75° C) 85° D) 95°

D5. Two figures are similar with scale factor 3. A 5° angle in the smaller figure corresponds to an angle of what measure in the larger figure?
A) 15° B) 5° C) 45° D) 3°

D6. (SPR) The legs of a right triangle have lengths 5 and 12. What is the length of the hypotenuse?

D7. An isosceles right triangle has a leg of length 5. What is the length of the hypotenuse?
A) 5 B) $5\sqrt{2}$ C) $5\sqrt{3}$ D) 10

D8. In a 30-60-90 triangle the hypotenuse has length 14. What is the length of the shorter leg?
A) 7 B) $7\sqrt{3}$ C) 14 D) $14\sqrt{3}$

D9. In a right triangle, the side opposite θ has length 3 and the hypotenuse has length 5. What is $\sin\theta$?
A) $\frac{3}{4}$ B) $\frac{3}{5}$ C) $\frac{4}{5}$ D) $\frac{4}{3}$

D10. Which expression is equal to $\cos 20^\circ$?
A) $\sin 20^\circ$ B) $\sin 70^\circ$ C) $\tan 20^\circ$ D) $\cos 70^\circ$

D11. (SPR) A circle has diameter 10. The circumference is $k\pi$. What is the value of k?

D12. A circle has radius 6. What is the length of an arc with central angle 60°?
A) π B) 2π C) 6π D) 12π

D13. The graph of $(x-2)^2+(y+1)^2=16$ in the xy-plane is a circle. What is the radius of the circle?
A) 2 B) 4 C) 8 D) 16

D14. A tree 10 feet tall casts a 5-foot shadow. At the same time, a nearby tree casts a 2-foot shadow. What is the height, in feet, of the second tree?

A) 2 B) 4 C) 7 D) 25

D15. A rectangle has side lengths 8 and 6. What is the length of a diagonal?

A) 7 B) 10 C) 14 D) 48

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7. Mixed module (SAT shape)

22 questions. Target **35 minutes**. Calculator / Desmos allowed. About the same mix as a real Math module: more Algebra and Advanced Math, fewer PSDA and Geometry, some SPR.

M1. $f(x) = 2x - 5$. What is $f(4)$?
A) 1 B) 3 C) 8 D) 13

M2. Which equation represents a line with slope $-\frac{1}{2}$ that passes through $(0, 6)$?
A) $y = 2x + 6$ B) $y = -\frac{1}{2}x + 6$ C) $y = -\frac{1}{2}x - 6$ D) $y = \frac{1}{2}x + 6$

M3. (SPR) A store charges \$8 per notebook plus a \$3 bag fee. The total is \$35. How many notebooks were bought?

M4. The system $y = x + 1$ and $2x + y = 10$ has solution (x, y). What is x?
A) 3 B) 4 C) 5 D) 9

M5. How many solutions does $3x - 6y = 9$ and $x - 2y = 3$ have?
A) Zero B) One C) Two D) Infinitely many

M6. Which point is a solution of $y > 2x - 1$?
A) $(2, 2)$ B) $(1, 2)$ C) $(3, 4)$ D) $(0, -2)$

M7. Which expression is equivalent to $4x(x - 3) - 2(x - 3)$?
A) $2(x - 3)(2x - 1)$ B) $2(x - 3)(2x + 1)$ C) $4x^2 - 10x + 6$ D) $2(x - 3)(2x - 1) + 1$

M8. (SPR) What is the minimum value of $y = x^2 - 4x - 1$?

M9. The function h is defined by $h(t) = -16t^2 + 48t + 4$. What is the maximum value of $h(t)$?
A) 4 B) 16 C) 36 D) 40

M10. Which equation models a quantity that starts at 20 and decays by 15% each year?
A) $y = 20 - 15x$ B) $y = 20(0.15)^x$ C) $y = 20(0.85)^x$ D) $y = 20(1.15)^x$

M11. (SPR) What is the larger solution of $|2x - 3| = 7$?

M12. How many distinct real solutions does $\frac{x}{x - 1} = \frac{2}{x - 1} + 1$ have?
A) Zero B) One C) Two D) Infinitely many

M13. (SPR) What is the valid solution of $\sqrt{3x - 2} = 4$?

M14. In a group, 25% of the students play piano. Of those who play piano, 40% also play violin. What percent of all the students play both?
A) 10% B) 15% C) 65% D) 40%

M15. (SPR) A jacket costs \$60. It is discounted 15%, then 8% tax is added to the **sale** price. What is the final price, in dollars? (Enter the exact decimal.)

M16. A poll is 61% with margin of error 5%. Which statement is **not** justified?

A) A reasonable interval for the true percent is 56% to 66%.

B) More than half of the population might favor the idea.

C) Exactly 61% of the whole population favors the idea.

D) A larger random sample would likely give a smaller margin of error.

M17. (SPR) Two printers run together at 40 and 55 pages per minute. How many minutes do they need to print 600 pages? Enter a decimal or a fraction.

M18. The angles of a triangle are x, $2x$, and $3x$. What is x, in degrees?

A) 20 B) 30 C) 36 D) 60

M19. (SPR) A 30-60-90 triangle has short leg 6. What is the length of the hypotenuse?

M20. A right triangle has legs 9 and 12. What is the hypotenuse?

A) 13 B) 15 C) 21 D) 108

M21. A circle has radius 8 and a central angle of 90°. What is the arc length?

A) 2π B) 4π C) 8π D) 16π

M22. In the xy-plane, the circle $(x+3)^2+(y-1)^2=9$ has center

A) $(3,-1)$ B) $(-3,1)$ C) $(3,1)$ D) $(-3,-1)$

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8. Answer keys

Instructor side.

Algebra

A1. $f(2) = 9$, $g(2) = 14$, $4 \cdot 9 - 14 = 22$. **C**
A2. $2y = -5x + 10$, $y = -\frac{5}{2}x + 5$. **A**
A3. 12 is dollars per extra shirt. **B**
A4. **B** $40 + 25m = 165$
A5. Add: $3x = 15$, $x = 5$, $y = 5$. **A**
A6. $a + c = 15$, $12a + 7c = 145 \Rightarrow a = 8$
A7. Second $\times 2$ is $2x - 4y = 10$, not 6. Parallel, distinct. **A** (zero)
A8. $-2x > 8 \Rightarrow x < -4$. **B**
A9. Only $(1, 0.5)$ satisfies both. **C**
A10. $m = \frac{8}{4} = 2$, $y - 5 = 2(x - 2)$, $y = 2x + 1$. **A**
A11. Perpendicular slope $-\frac{1}{3}$, through $(0, 4)$. **B**
A12. $2(5x - 2) = 4(x + 6) \Rightarrow x = \frac{14}{3}$
A13. $120 - 4t = 0 \Rightarrow t = 30$
A14. $6 + k > 10 \Rightarrow k > 4$. **A**
A15. $7 - 6x + 3 = 2x + 8 \Rightarrow x = \frac{1}{4}$. **A**
A16. $0.10t = 12 \Rightarrow t = 120$

Advanced Math

B1. Factor $x + 4$: $(x + 4)(2x - 1)$. **A**
B2. $\frac{(x-3)(x+3)}{x-3} = x + 3$. **B**
B3. $9x^4y^2 \Rightarrow 9 + 4 + 2 = 15$
B4. Difference of cubes. **B**
B5. $x = 3$ or $x = -1$. **A**
B6. $g(x) = (x - 0)^2 + 55$. Minimum **55**. **C**
B7. $x = -\frac{b}{2a} = 4$
B8. 9

B9. $x = 3$ or $x = -\frac{11}{3}$. **B**

B10. Absolute value cannot equal a negative. **A**

B11. $x + 3 = 2x - 2 \Rightarrow x = 5$

B12. Candidates 3 and -1; $\sqrt{1} \neq -1$. **B**

B13. $2^{x+1} = 2^4 \Rightarrow x = 3$. **B**

B14. $200 \times (1.5)^2 = 450$

B15. $x^2 - 2x - 3 = 0 \Rightarrow$ two points. **C**

B16. $3 \cdot 16 = 48$

PSDA

C1. 30

C2. 150

C3. $0.30 \times 0.40 = 0.12$. **B**

C4. $80 \times 1.25 = 100$, $100 \times 0.75 = 75$. **B**

C5. 40

C6. $800 \times 1.21 = 968$. **B**

C7. $72/1.2 = 60$

C8. Middle value **3**. **B**

C9. Sum must be 400; known sum 320; fifth **80**

C10. Seniors only: $18/30$. **B**

C11. **B**

C12. **B**

C13. $40/8 = 5$. **A**

C14. Over the grand total: $18/50$. **B**

Geometry

D1. 27

D2. $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$. **C**

D3. Area ratio $4:9$, $20 \times \frac{9}{4} = 45$. **C**

D4. 75°. **B**

D5. Angles do not scale. **B**

D6. 13

D7. $5\sqrt{2}$. **B**

D8. Half the hypotenuse. **A**

D9. $\frac{3}{5}$. **B**

D10. $\cos\theta = \sin(90^\circ - \theta)$. **B**

D11. 10

D12. $\frac{1}{6} \cdot 12\pi = 2\pi$. **B**

D13. $r = 4$. **B**

D14. $\frac{10}{5} = \frac{h}{2} \Rightarrow h = 4$. **B**

D15. 10. **B**

Mixed module

M1. 3. **B**

M2. **B**

M3. $8n + 3 = 35 \Rightarrow n = 4$

M4. $2x + (x + 1) = 10 \Rightarrow x = 3$. **A**

M5. Second $\times 3$ is the first equation. Same line. **D**

M6. $2 > 1$ only for $(1, 2)$. **B**

M7. $(x - 3)(4x - 2) = 2(x - 3)(2x - 1)$. **A**

M8. Vertex at $x = 2, y = -5$

M9. $t = 1.5$, $h(1.5) = 40$. **D**

M10. Multiply by 0.85 each year. **C**

M11. $x = 5$ (the other is -2)

M12. Contradiction after clearing; $x = 1$ forbidden. **A**

M13. $3x - 2 = 16 \Rightarrow x = 6$

M14. $0.25 \times 0.40 = 0.10$. **A**

M15. $60 \times 0.85 = 51$, $51 \times 1.08 = 55.08$

M16. **C**

M17. $600/95 = \frac{120}{19}$ (about 6.32)

M18. $6x = 180 \Rightarrow x = 30$. **B**

M19. 12

M20. 9-12-15. **B**

M21. $\frac{1}{4} \cdot 16\pi = 4\pi$. **B**

M22. Center $(-3, 1)$. **B**